

5 Calibration

5.3 Calibration scale and correct axis position

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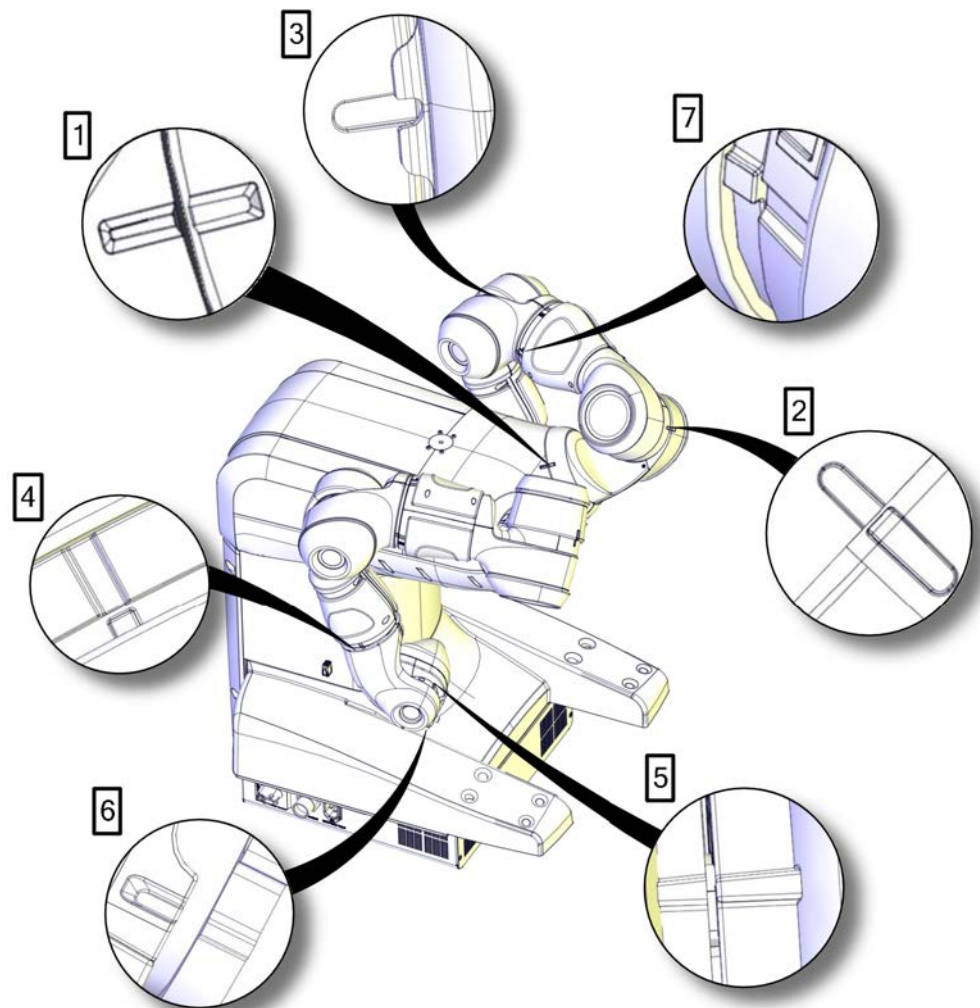
Introduction

This section specifies the calibration scale positions and/or correct axis positions.

Calibration scales/marks

This illustration shows the positions of the calibration scales and marks on the robot.

The number next to the enlargement corresponds to the axis number.



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5.7 Updating revolution counters

Introduction

This section describes how to do a rough calibration of each robot axis, which updates the revolution counter value for each axis using the FlexPendant.

The procedure can be summarized accordingly:

- 1 Manually move the manipulator to the calibration position.
- 2 Select the Calibration with hall sensors (CalHall) routine.
- 3 Select the function Update of revolution counters.
- 4 Store the revolution counter setting.

Each step is described in detail in following sections.

Step 1 - Manually moving the manipulator to the calibration position

	Action	Note
1	 CAUTION When releasing the holding brakes, the robot axes may move very quickly and sometimes in unexpected ways!	
2	Release the brakes of the robot arm to be calibrated and move the arm manually so that the synchronization mark of each joint is aligned. The robot now stands in its calibration position.	The synchronization marks are shown in Calibration scale and correct axis position on page 400 . There is a tolerance for the joint position. The edge of a mark should be at least within the area of the opposite mark.

Step 2 - Selecting the Calibration with hall sensors (CalHall) routine

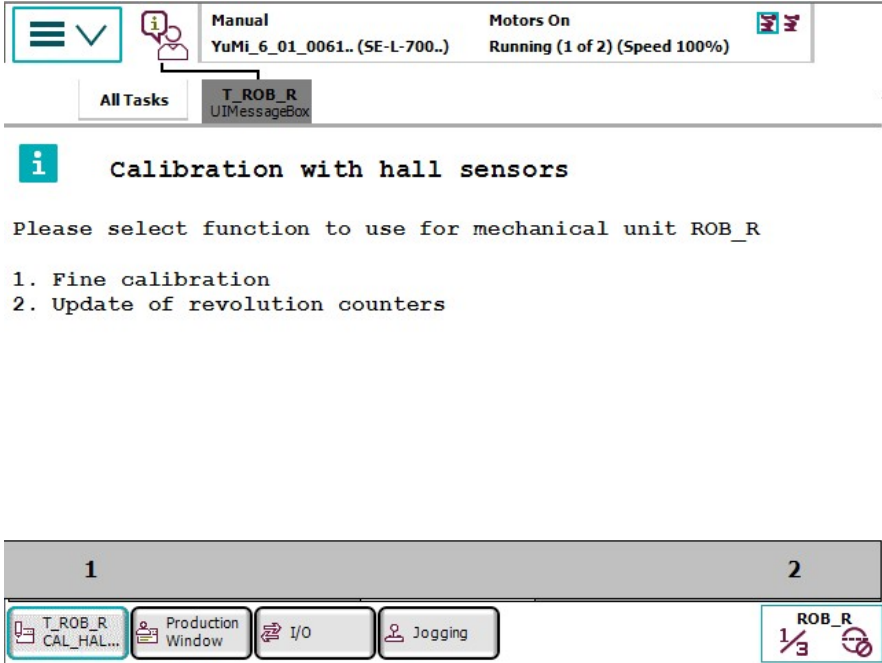
	Action	Note
1	Open the Program Editor on the FlexPendant.	
2	Select the task that corresponds to the robot arm to be calibrated. Tap Open .	
3	If necessary, create a new program. This needs to be done if no existing program is available.	
4	Select Debug and tap PP to Main .	
5	Select Debug and tap Call Routine...	
6	Select CalHall .	
7	Go to Motor On and press the Start button .	

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Step 3 - Selecting the function Update of revolution counters

	Action
1	In the calibration with hall sensors routine (CalHall), tap 2 to select the function of updating the revolution counters.  xx1400002694

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Action
2 Tap to select which joint(s) to update revolution counters for. Joints 1, 2, 3 and 4 are selectable in the first window. Tap Next to open the second window where joints 5, 6 and 7 are selectable. Manual YuMi_6_01_0061.. (SE-L-700..) Motors On Running (1 of 2) (Speed 100%) All Tasks T_ROB_R U:\MessageBox Selection of joint(s) to update Choose joint(s) to update revolution counters for ROB_R 1. ☒ [X] 2. ☒ [X] 3. ☒ [X] 4. ☒ [X] 5. ☐ [X] 6. ☐ [X] 7. ☐ [X] 1 2 3 4 Next Production Window T_ROB_R CAL_HA... ROB_R 1/3 xx1400002695 Manual YuMi_6_01_0061.. (SE-L-700..) Motors On Running (1 of 2) (Speed 100%) All Tasks T_ROB_R U:\MessageBox Selection of joint(s) to update Choose joint(s) to update revolution counters for ROB_R 1. ☒ [X] 2. ☒ [X] 3. ☒ [X] 4. ☒ [X] 5. ☒ [X] 6. ☒ [X] 7. ☒ [X] 5 6 7 Next Production Window T_ROB_R CAL_HA... I/O ROB_R 1/3 xx1400002696

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Step 4 - Storing the revolution counter setting

Action
1 Tap OK to start the sampling procedure of the hall sensor. One at a time, each selected joint now rotates back and forth several times to calculate and find the exact position where the hall sensor is aligned with the magnet. Calibration order: axis 1-2-3-4-5-6-7. Note This procedure may take several minutes. Manual YuMi_6_01_0061.. (SE-L-700..) Motors On Running (1 of 2) (Speed 100%) All Tasks T_ROB_R UIMessageBox Sampling of Hall sensor starting This operation may take several minutes to complete. Press OK to start! T_ROB_R CAL_HAL... Production Window OK T_ROB_R CAL_HAL... Production Window ROB_R 1/3 xx1400002697

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2 When the sampling of hall sensors is finished, tap OK to confirm.

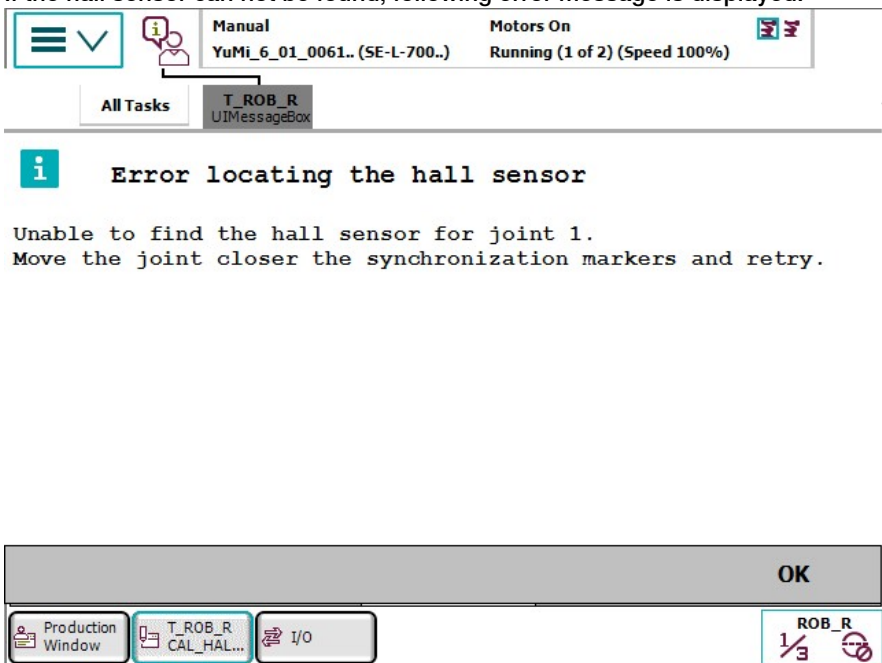
The screenshot shows the T_ROB_R control interface. At the top, there is a status bar with a menu icon, a checkmark, and a notification icon. Below this, the text "Manual YuMi_6_01_0061.. (SE-L-700..)" and "Motors On Running (1 of 2) (Speed 100%)" are displayed. A button labeled "All Tasks" is visible. The main area shows a message: "Sampling of Hall sensor finished" with an information icon. Below the message, it says "Press OK to confirm!". At the bottom, there is a large "OK" button. The bottom status bar shows "T_ROB_R CAL_HAL..." and "ROB_R 1/3" with a circular arrow icon.

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Action	
3	If the hall sensor can not be found, following error message is displayed.  xx1400002698 There can be several causes for this error. • The joints are not sufficiently aligned according to the synchronization marks. Move the joints so that the synchronization marks are aligned. Tap OK and restart the fine calibration procedure. The joints prior to the joint that stopped the calibration procedure are calibrated and do not need to be calibrated again. Joints coming after the joint that stopped the calibration procedure has not been calibrated. (Calibration order: axis 1-2-3-4-5-6-7.) • The hall sensor is defect. Perform troubleshooting of I/O signals. Possible replacement of hall sensor may be required.

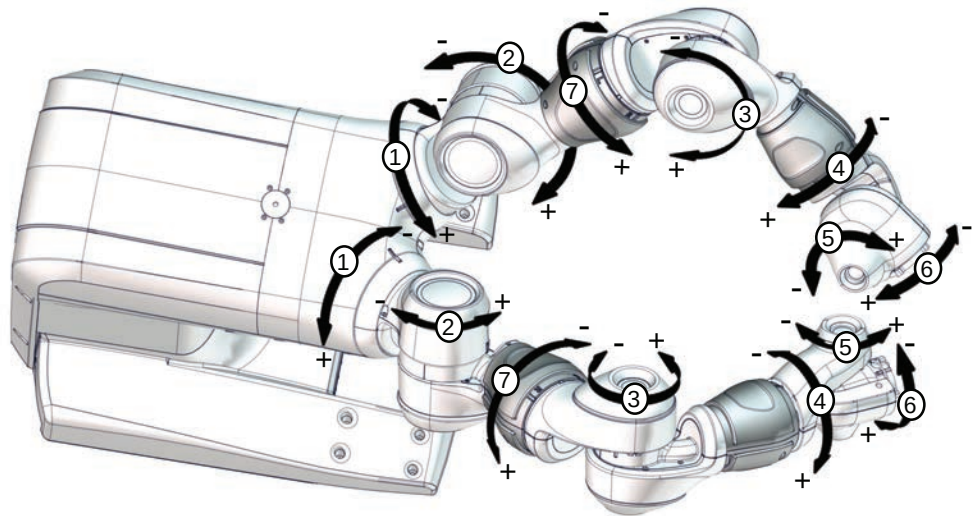
5.8 Calibration movement directions for all axes

Overview

When calibrating, the axis must consistently be run towards the calibration position in the same direction in order to avoid position errors caused by backlash in gears and so on. Positive directions are shown in the graphic below.

This is normally handled by the robot calibration software.

Calibration movement directions, 7 axes



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5 Calibration

5.9 Verifying the calibration position

5.9 Verifying the calibration position

Introduction

Verify the calibration position of the robot before beginning any programming of the robot system. This may be done:

- Using a MoveAbsJ instruction with argument according to calibration position degrees on all axes.
- Using the **Jogging** window on the FlexPendant.

Using a MoveAbsJ instruction

Use this procedure to create a program that runs all the robot axes to their calibration position.

	Action	Note
1	On ABB menu tap Program editor .	
2	Create a new program.	
3	Use MoveAbsJ in the Motion&Proc menu.	
4	Create the following program for the right arm: MoveAbsJ [[0, -130, 30, 0, 40, 0], [-135, 9E9, 9E9, 9E9, 9E9, 9E9]] NoEOffs, v1000, fine, tool0; Create the following program for the left arm: MoveAbsJ [[0, -130, 30, 0, 40, 0], [135, 9E9, 9E9, 9E9, 9E9, 9E9]] NoEOffs, v1000, fine, tool0;	
5	Run the program in manual mode.	
6	Verify that the calibration marks for the axes align correctly. If they do not, then update the revolution counters.	See Calibration scale and correct axis position on page 400 and Updating revolution counters on page 415 .

Using the jogging window

Use this procedure to jog the robot to the calibration position for all axes.

	Action	Note
1	On the ABB menu, tap Jogging .	
2	Tap Motion mode to select group of axes to jog.	
3	Tap to select the axis to jog, axis 1, 2, or 3.	
4	Manually run the robots axes to a position where the axis position value read on the FlexPendant, is equal to the calibration position degrees.	Degrees are specified in Exact axis positions in degrees on page 401 .
5	Verify that the calibration marks for the axes align correctly. If they do not, then update the revolution counters!	See Calibration scale and correct axis position on page 400 and Updating revolution counters on page 415 .